A Measure-Zero Periodic Set Containing Every Rational Parameter

Here is a particularly sharp warning against cardinality and measure arguments. It is a genuine analytic, time-reversible Hamiltonian brake system—the same broad dynamical language as the Burrau problem.

For $0 < u < 1$, consider the two-degree-of-freedom Hamiltonian

$$H_u(x, y, p_x, p_y) = \frac{1}{2}\big(p_x^2 + x^2\big) + \frac{1}{2}\big(p_y^2 + u^2 y^2\big),$$

and begin at the brake point

$$x(0) = y(0) = 1, \qquad p_x(0) = p_y(0) = 0.$$

Hamilton's equations give

$$x(t) = \cos t, \qquad y(t) = \cos(ut),$$

and

$$p_x(t) = -\sin t, \qquad p_y(t) = -u\sin(ut).$$

The second-brake condition

A second brake occurs precisely when both momenta vanish. Thus there must be positive integers m, n for which

$$t = n\pi, \qquad ut = m\pi.$$

Eliminating t gives

$$u = \frac{m}{n} \in \mathbb{Q}.$$

Conversely, if $u = m/n$, then $t = n\pi$ is a second-brake time. Time reversal about that brake point produces a periodic orbit. Hence

$$\boxed{\text{a second brake exists exactly when } u \in \mathbb{Q}}$$

and therefore

$$\boxed{\text{every rational parameter is periodic, while every irrational parameter is nonperiodic.}}$$

Why this is surprising

The periodic-parameter set $\mathbb{Q} \cap (0, 1)$

- is countable;
- has Lebesgue measure zero;

- contains no interval;
- is surrounded arbitrarily closely by nonperiodic parameters;
- nevertheless contains every rational parameter, including every rational Euclid parameter used to generate a Pythagorean triple.

For irrational u, the trajectory winds densely on its invariant two-torus. For rational u, the two frequencies are commensurable and the trajectory closes.

> In this analytic Hamiltonian brake problem, almost every parameter is nonperiodic, yet every single rational parameter is periodic.

Thus neither "periodic parameters have measure zero" nor "almost every parameter is nonperiodic" can settle the Pythagorean–Burrau conjecture. If the Burrau brake map behaved like this elementary oscillator family, the conjecture would be maximally false even though its periodic parameters still formed a measure-zero set. An exact structural or arithmetic obstruction is therefore indispensable.